1 Introduction

- Most mathematical models of real-life processes involve differential equations (e.g. mechanical systems, electrical circuits, chemical reactions, biological populations, financial markets).
- An ordinary differential equation (ODE) is an equation relating values of a function ($y(t)$, say) to its derivatives, for example,

$$\frac{dy}{dt} + 6y = 0 \quad \text{or} \quad \frac{d^2y}{dt^2} - \frac{4t+1}{t}\frac{dy}{dt} + \frac{2(2t+1)}{t}y = 4e^{2t}\,.$$

- The function $y(t)$ is the **unknown function** that is to be found by solving the differential equation. Recall that a function $y(t)$ has an **independent variable** t and a **dependent variable** y.
- A differentiable[1] function $y(t)$ is a **solution of a differential equation** in an interval $I : a < t < b$, say, if the substitution of $y(t)$ into the differential equation produces an equality which is true for all $t \in I$. The interval I (which must be continuous) may be finite or infinite, i.e. $a \to -\infty$ and $b \to \infty$ are allowed.

[1] A differentiable function $y(t)$ is a function whose derivative

$$\frac{dy}{dt} = \lim_{h\to 0}\frac{y(t+h)-y(t)}{h}$$

is defined at all points in the domain. Note that $y(t)$ must be defined in order that dy/dt is defined. See the MTH1002, part 1, lecture notes.

Example 1.1 Consider a bank account in which the money grows in such a way that the increase per unit time t is proportional to the current balance $y(t)$, with a constant coefficient k, i.e. the amount of money in the account satisfies the ODE

$$\frac{dy}{dt} = ky\,.$$

Verify that is $y = C\exp(kt)$ is a solution for $-\infty < t < \infty$.

- A **partial differential equation** (PDE) is an equation that relates values of a function of more than one variable (e.g. $y(x,t)$, say) to its partial derivatives, for example

$$\frac{\partial^2 y}{\partial x^2} + \frac{\partial^2 y}{\partial t \partial x} + 3\frac{\partial y}{\partial t} + xy = t^3 .$$

  Here $y(x,t)$ is the unknown function and x and t are the independent variables.

- The **order of a differential equation** is the order of the highest derivative of the unknown function.

Example 1.2 What is order of the following differential equations?

1. $\frac{dy}{dt} + 6y = 0$
2. $\frac{\partial^3 y}{\partial x^3} + \frac{\partial^2 y}{\partial t^2} + 3xy^6 = \frac{\partial^4 f}{\partial x^4}$, where f is a given function of x

- The nth order ordinary differential equation

$$\frac{d^n y}{dt^n} = f\left(t, y, \frac{dy}{dt}, \ldots, \frac{d^{(n-1)}y}{dt^{(n-1)}}\right)$$

  is said to be **linear** if f is a linear function of the variables $y, \frac{dy}{dt}, \ldots, \frac{d^{(n-1)}y}{dt^{(n-1)}}$, otherwise the differential equation is said to be **nonlinear**.

Example 1.3 Are the following differential equations linear or nonlinear?

1. $a_0(t)\frac{d^n y}{dt^n} + a_1(t)\frac{d^{n-1}y}{dt^{n-1}} + \cdots + a_n(t)y = g(t)$
2. $\frac{d^3 y}{dt^3} + 2e^t\frac{d^2 y}{dt^2} + y\frac{dy}{dt} = t^5$
3. $\frac{d^2\theta}{dt^2} + \frac{g}{L}\sin\theta = 0$, where g and L are constants
4. $\frac{d^2\theta}{dt^2} + \frac{g}{L}\theta = 0$, where g and L are constants

2 First Order Ordinary Differential Equations

Reference: Boyce & DiPrima, Chapter 2

In this section we consider differential equations in the (standard) form

$$\boxed{\frac{dy}{dt} = f(t,y)\,,} \tag{2.1}$$

where f (the right-hand side) is a given function of the two variables t and y. Any differentiable function $y = \phi(t)$ that satisfies this equation for all t in some interval $I : \alpha < t < \beta$ is called a solution. Our objective is discuss whether such functions exist and if so to find the solution. There are different methods of finding the solution for different forms of the function f.

2.1 Linear Equations

When the function f depends linearly on the dependent variable y then equation (2.1) can be written in the form

$$\boxed{\frac{dy}{dt} + p(t)y = g(t)\,,} \tag{2.2}$$

where we assume that p and g are given continuous functions on the interval $I : \alpha < t < \beta$. The equation is known as a **first order linear differential equation**.

2.1.1 Integrating factor method[2]

[2] Integrating factors have been seen before in MTH1002, part 2

To solve equation (2.2) we first find the **integrating factor** $\mu(t)$ defined by

$$\boxed{\mu(t) = \exp\left(\int p(t)\,dt\right)\,.} \tag{2.3}$$

Note that $\mu(t)$ is positive for all t.

Then we note that differentiating $\mu(t)$ gives

$$\frac{d\mu}{dt} = \exp\left(\int p(t)\,dt\right)p(t) = \mu(t)p(t)\,. \tag{2.4}$$

We will use this below.

Now multiply (2.2) by $\mu(t)$ to obtain

$$\mu(t)\frac{dy}{dt} + \mu(t)p(t)y = \mu(t)g(t)\,. \tag{2.5}$$

and notice that this is the same equation as

$$\boxed{\frac{d}{dt}\left[\mu(t)y\right] = \mu(t)g(t)} \tag{2.6}$$

because we can expand the left-hand side of (2.6) using the product rule

$$\begin{aligned}\frac{d}{dt}\left[\mu(t)y\right] &= \mu(t)\frac{dy}{dt} + y\frac{d\mu}{dt}\\ &= \mu(t)\frac{dy}{dt} + y\mu(t)p(t) \quad \text{using (2.4).}\end{aligned}$$

Now we solve (2.6) by integrating both sides to obtain

$$\mu(t)y = \int \mu(t)g(t)\,dt + c\,,$$

where c is an arbitrary constant. Rearranging gives the **general solution**

$$y = \frac{\int \mu(t)g(t)\,dt + c}{\mu(t)}\,. \tag{2.7}$$

The graphs of $y(t)$ for a chosen constant c are called **integral curves**. They are the solutions of the first order linear differential equation (2.2). There is a infinite family of such curves, one for each c.

A first order differential equation together with an **initial condition**, written

$$\boxed{y(t_0) = y_0\,,}$$

is known as an **initial value problem** (IVP). The solution to an initial value problem is the one member of the infinite family of integral curves that passes through the point (t_0, y_0).

Example 2.1 Consider the initial value problem

$$\frac{dy}{dt} - 2y = e^{2t}, \quad y(0) = 2\,.$$

Plot the integral curves and find the solution of the IVP.

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2.2 Separable Equations[3]

[3] Separable equations have been seen before in MTH1002, part 2

Recall the first order ODE (2.1)

$$\frac{dy}{dt} = f(t,y)\,.$$

If $f(t,y)$ can be separated into a function of t only ($F(t)$, say) multiplied by a function of y only ($G(y)$, say) then we have a **separable first order ODE**

$$\boxed{\frac{dy}{dt} = F(t)G(y(t))} \tag{2.8}$$

2.2.1 Special solutions

If $G(y_*) = 0$ then $y(t) = y_*$ is a **special solution**.

2.2.2 Typical solutions

If $G \neq 0$ then we may divide by it and find the **typical solutions** of

$$\frac{1}{G(y)}\frac{dy}{dt} = F(t).$$

Integrating with respect to t gives

$$\int \frac{1}{G(y)}\frac{dy}{dt}dt = \int F(t)dt + c,$$

where c is a constant. Now we take y as the integration variable on the left-hand side to obtain

$$\int \frac{1}{G(y)}dy = \int F(t)dt + c.$$

2.2.3 General solution

The general solution is the union of the set of all solutions.

Example 2.2 Find the solution of the initial value problem

$$\frac{dy}{dt} = \frac{y\cos t}{1+2y^2}, \quad y(0) = 1\,.$$

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2.2.4 Test for separability

Note that if $f(t,y) = F(t)G(y)$ for all t and y then

$$f(a,b)f(c,d) = F(a)G(b)F(c)G(d) = f(a,d)f(c,b)$$

for all a, b, c and d. Hence, if you can find one set of a, b, c and d for which this equality does not hold, then you know that the ODE is not separable.

Activity 2.3 Is the ODE

$$y^2\frac{dy}{dt} + t^2 = ty^2$$

separable?

2.3 Homogeneous Equations

The first order ODE (2.1)

$$\frac{dy}{dt} = f(t,y)\,,$$

is said to be homogeneous whenever the function f does not depend on t and y separately, but only depends on their ratio y/t or t/y. Thus, homogeneous first order ODEs take the form

$$\boxed{\frac{dy}{dt} = H\left(\frac{y}{t}\right)} \tag{2.9}$$

The form of equation (2.9) suggests that the equation can be simplified by introducing a new variable to represent the ratio. Thus we set

$$v = \frac{y}{t}\,, \quad t \neq 0\,.$$

Then, using the product rule,

$$y = vt \quad \Rightarrow \frac{dy}{dt} = v + t\frac{dv}{dt}$$

and so substituting into (2.9) gives the ODE

$$v + t\frac{dv}{dt} = H(v)$$

which is of separable form

$$\int \frac{dt}{t} = \int \frac{dv}{H(v) - v}$$

for $H(v) \neq v$.

Example 2.4 Solve the differential equation

$$t\frac{dy}{dt} = t + y\,.$$

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2.3.1 Test for homogeneity

A homogeneous equation has

$$f(wt, wy) = f(t, y), \quad \forall\, w, t, y\,.$$

Hence, if you can find a set of w, t, y for which this equality does not hold, then you know that the ODE is not homogeneous.

Important note: The classification 'homogeneous' is used in more than one way in the study of differential equations. With experience, you will be able to distinguish between these.

2.4 Existence & Uniqueness

In the previous sections we have used a variety of methods to find solutions to the initial value problem

$$\frac{dy}{dt} = f(t,y), \quad y(t_0) = y_0\,. \tag{2.10}$$

In this section we address questions of a more general nature, such as:

- Does an initial value problem always have a solution?
- Can it have more than one solution?
- Is the solution valid for all t, or only for some restricted interval near the initial point t_0?

First, we require the definition of a continuous function:[4]

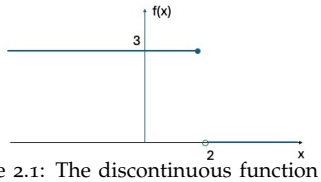


Figure 2.1: The discontinuous function $f(x) = 3$ for $x \leq 2$ and $f(x) = 0$ for $x > 2$.

[4] see the lecture notes for MTH1002, part 1. Note that if the left-hand and right-hand limits are different

$$\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x)$$

then

$$\lim_{x\to a} f(x) \quad \text{does not exist.}$$

For example, the function defined by

$$f(x) = \begin{cases} 3 & x \leq 2 \\ 0 & x > 2 \end{cases}$$

is not continuous at $x = 2$ (see figure 2.1). Also, if a function is not defined at a certain point then it cannot be continuous at that point. For example,

$$f(x) = \frac{1}{x-3}$$

is not defined at $x = 3$, see figure 2.2.

Definition 2.5 A function $f(x)$ is **continuous** at $x = a$ if

$$\lim_{x\to a} f(x) = f(a)\,.$$

A function is continuous on an interval if it is continuous at all points in that interval.

2.4.1 Linear first order ODEs

Theorem 2.6 If the functions $p(t)$ and $g(t)$ are continuous on the open interval $I : \alpha < t < \beta$, with $t_0 \in I$, then there exists a unique function $y = \phi(t)$ that satisfies the linear differential equation

$$\frac{dy}{dt} + p(t)y = g(t) \tag{2.11}$$

for all t in I, with $y(t_0) = y_0$, where y_0 is an arbitrary prescribed value.

Note that this theorem asserts both existence and uniqueness of the solution of the initial value problem (i.e. we are told that the IVP has a solution and it has only one solution). The theorem also states that the solution exists throughout the interval I on which p and g are continuous and that contains t_0.

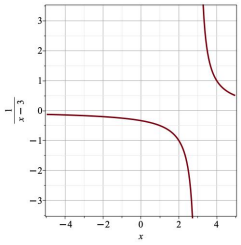


Figure 2.2: The discontinuous function $f(x) = 1/(x-3)$ that is undefined at $x = 3$.

2.4.2 First order ODEs

Definition 2.7 In one-dimensional space, a **neighbourhood** of a point $t \in \mathbb{R}$ is any (possibly very small) open interval containing the point t. Similarly, in two-dimensional space, a neighbourhood of a point $(t, y) \in \mathbb{R}^2$ is any open set (e.g. a rectangle or a disc) containing the point (t, y). A certain property of a function f is said to be a **local property**, or is **locally satisfied**, if it is satisfied in a neighbourhood of the point.

Theorem 2.8 — Peano's existence theorem. Let the function $f(t, y)$ be continuous in a neighbourhood of (t_0, y_0). Then the IVP

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

is **solvable locally**, i.e. it has **at least one solution** in a neighbourhood of t_0.

Note that the interval of t where the solution exists may be smaller than the interval over which f is continuous.

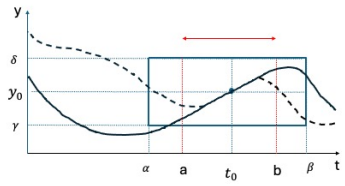


Figure 2.3: Peano's existence interval.

Theorem 2.9 — Picard's existence & uniqueness theorem. Let the functions $f(t, y)$ and its partial derivative $\partial f/\partial y$ be continuous in a neighbourhood of (t_0, y_0). Then the IVP

$$\frac{dy}{dt} = f(t, y), \quad y(t_0) = y_0$$

is **uniquely solvable locally**, i.e. there is **exactly one solution** in a neighbourhood of t_0.

Note that the interval of t over which the solution is unique may be smaller than the interval over which f and $\partial f/\partial y$ are continuous.

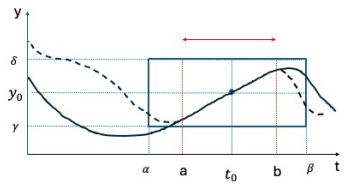


Figure 2.4: Picard's uniqueness interval.

Example 2.10 Without solving the IVP

$$t\frac{dy}{dt} + 2y = 4t^2, \quad y(1) = 2,$$

find the interval in which it has a unique solution. Then solve the IVP to find the solution in that interval.

.

Example 2.11 Find the interval on which the IVP

$$\frac{dy}{dt} + (\tan t)y = \sin t, \quad y(\pi) = 0$$

is guaranteed to have a unique solution.

We note that questions concerning existence and uniqueness of solutions are different to the question of actually finding a solution. One may ask why we are interested in existence and uniqueness. One reason is that in a mathematical modelling problem you may find a solution but it might not correspond to the actual behaviour of the real physical system if the solution isn't unique; i.e. you have found a solution but not the one of physical interest. Alternatively, if you can show that the solution exists and is unique, then, even if you cannot find the solution analytically, you can proceed to find an approximation to it computationally.

3 Second Order Linear Ordinary Differential Equations: Initial Value Problems

Reference: Boyce & DiPrima, Chapter 3

Definition 3.1 A **second order linear ordinary differential equation** has the form

$$a_2(t)y'' + a_1(t)y' + a_0(t)y = G(t)\,, \tag{3.1}$$

where we have used a prime symbol $'$ to denote a derivative. The coefficients on the left-hand side (namely $a_2(t)$, $a_1(t)$ and $a_0(t)$), and the function $G(t)$ on the right-hand side, are specified (known) functions of the independent variable t. This is known as the **standard form** of a linear second order ODE. Alternatively, very often we deal with the equation in the **normalised form**

$$y'' + p(t)y' + q(t)y = g(t)\,. \tag{3.2}$$

Definition 3.2 The second order linear ordinary differential equation (3.2) is said to be **homogeneous** if the right-hand side $g(t)$ is zero for all t. Otherwise the equation is called **nonhomogeneous**. The term $g(t)$ is sometimes referred to as the nonhomogeneous term.

3.1 Fundamental set of solutions of linear homogeneous equations

In this section we set $g(t)$ to zero and we briefly describe some general theory for the second order linear homogeneous ODE

$$y'' + p(t)y' + q(t)y = 0\,. \tag{3.3}$$

When we have understood how to solve this homogeneous equation, we will be able to make progress in solving a nonhomogeneous equation.

Theorem 3.3 — Principle of superposition for homogeneous equations. If $y_1(t)$ and $y_2(t)$ are two solutions of the homogeneous equation (3.3) then

$$y_3(t) = c_1y_1(t) + c_2y_2(t)$$

is a solution of (3.3) for any constants c_1 and c_2.

Proof.

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Definition 3.4 The Wronski determinant, or **Wronskian** for short, of two functions $y_1(t)$ and $y_2(t)$ is the function

$$\begin{aligned} W(t) &= \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} \\ &= y_1(t)y_2'(t) - y_2(t)y_1'(t) \\ &= W[y_1, y_2](t) \end{aligned} \tag{3.4}$$

Theorem 3.5 Let y_1 and y_2 be solutions of

$$y'' + p(t)y' + q(t)y = 0\,,$$

where p and q are continuous on an open interval $I : \alpha < t < \beta$. Then y_1 and y_2 are **linearly independent** on I if and only if $W[y_1, y_2](t)$ is never zero in I.

Theorem 3.6 — Abel's Identity. If y_1 and y_2 are solutions of

$$y'' + p(t)y' + q(t)y = 0\,,$$

where p and q are continuous on an open interval $I : \alpha < t < \beta$, then the Wronskian is given by

$$W[y_1, y_2](t) = c \exp\left[-\int p(t)\,dt\right], \tag{3.5}$$

where c is a certain constant that depends on y_1 and y_2, but not on t. Consequently, $W[y_1, y_2](t)$ is either zero for all t in I (if $c = 0$) or it is never zero in I (if $c \neq 0$).

Proof.

.

Theorem 3.7 If y_1 and y_2 are two linearly independent solutions of

$$y'' + p(t)y' + q(t)y = 0\,, \tag{3.6}$$

where p and q are continuous on an open interval $I : \alpha < t < \beta$, then the family of solutions

$$y = c_1 y_1(t) + c_2 y_2(t) \tag{3.7}$$

with arbitrary constant coefficients c_1 and c_2 includes every solution of (3.6) and we call (3.7) the **general solution** of (3.6). The solutions y_1 and y_2 are said to form a **fundamental set of solutions** of (3.6) on I.

Example 3.8 Verify that the functions $y_1(t) = \cos(3t)$ and $y_2(t) = \sin(3t)$ satisfy the ODE

$$y'' + 9y = 0\,.$$

Do they constitute a fundamental set of solutions? Justify your answer. Also verify Abel's identity.

.

3.2 Initial value problems

An **initial value problem** (IVP) for a second order linear differential equation comprises the differential equation and a pair of initial conditions

$$y(t_0) = y_0, \quad y'(t_0) = y_0', \tag{3.8}$$

where y_0 and y_0' are given numbers. We note that the initial conditions prescribe not only the particular point (t_0, y_0) through which the graph of the solution must pass, but also the slope y_0' of the graph at that point.

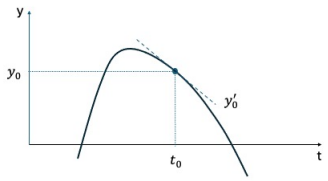


Figure 3.1: The initial condition.

Theorem 3.9 Consider the initial value problem

$$y'' + p(t)y' + q(t)y = g(t), \quad y(t_0) = y_0, \quad y'(t_0) = y_0', \tag{3.9}$$

where p, q and g are continuous on the open interval $I : \alpha < t < \beta$, with $t_0 \in I$. Then there exists a unique solution $y = \phi(t)$ of the IVP throughout the interval I.

Example 3.10 Without solving the IVP

$$(t^2 - 3t)y'' + t\,y' - (t+3)\,y = 0, \quad y(1) = 2, \quad y'(1) = 1\,,$$

find the longest interval in which the solution is guaranteed to exist.

Example 3.11 Without solving the IVP

$$(t-2)y'' + t\,y' + (\ln|t|)\,y = 0, \quad y(1) = 0, \quad y'(1) = 1\,,$$

find the longest interval in which there is certain to be a unique solution.

3.3 Homogeneous equations with constant coefficients[5]

[5] These ODEs have been seen before in the MTH1002, part 2, lecture notes.

Consider the second order ODE (3.1) in the homogeneous case (in which $G(t) \equiv 0$) and when the coefficients a_2, a_1 and a_0 are constants. In this case (3.1) can be written as

$$ay'' + by' + cy = 0\,, \tag{3.10}$$

with $a \neq 0$.

We postulate a solution of the form

$$y = e^{mt}\,, \tag{3.11}$$

where m is a constant that is to be determined. We calculate the derivatives

$$y' = me^{mt}, \quad y'' = m^2 e^{mt}$$

and substitute into (3.10) to obtain the **characteristic equation**

$$am^2 + bm + c = 0\,. \tag{3.12}$$

There are two roots of this characteristic equation:

$$m_{1,2} = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a}\,.$$

They are:

- Real and different when $b^2 - 4ac > 0$, giving $m_1 \neq m_2$.
- Complex conjugates when $b^2 - 4ac < 0$, giving $m_1 = \lambda + i\mu$ and $m_2 = \lambda - i\mu$.
- Real and repeated when $b^2 - 4ac = 0$, giving $m_1 = m_2$.

3.3.1 Real and different roots

The two solutions of (3.10) are

$$y_1(t) = e^{m_1 t}, \quad y_2(t) = e^{m_2 t}\,, \quad m_1 \neq m_2\,.$$

The Wronskian

$$W[y_1, y_2](t) = e^{m_1 t} m_2 e^{m_2 t} - e^{m_2 t} m_1 e^{m_1 t} = (m_2 - m_1) e^{(m_1 + m_2)t} \neq 0 \quad \forall t$$

since the roots are different, i.e. $m_1 \neq m_2$

Thus the solutions are linearly independent for all t and by Theorem 3.7 the **general solution** is

$$y = c_1 e^{m_1 t} + c_2 e^{m_2 t}, \quad t \in \mathbb{R}\,. \tag{3.13}$$

Activity 3.12 Verify that the linear sum (3.13) is a solution of (3.10) by differentiating and substituting.

Activity 3.13 Now suppose that we want to find a particular member of the family of solutions (3.13) that satisfies the initial conditions

$$y(t_0) = y_0, \quad y'(t_0) = y_0'.$$

Find c_1 and c_2.

Answer: $c_1 = \frac{y_0 m_2 - y_0'}{m_2 - m_1} \exp(-m_1 t_0)$, $c_2 = \frac{y_0' - y_0 m_1}{m_2 - m_1} \exp(-m_2 t_0)$

.

.

Example 3.14 Find the general solution of

$$y'' + 5y' + 6y = 0\,.$$

Example 3.15 Find the solution of the initial value problem

$$y'' + 5y' + 6y = 0\,, \quad y(0) = 2, \quad y'(0) = 3.$$

Graph the solution and describe its behaviour as t increases.

.

In summary, when the roots are real and different ($m_1 \neq m_2$) the general solution is a sum of exponentials. As t increases, the magnitude of the solution either tends to zero (when both exponents are negative), grows rapidly (when at least one exponent is positive), or approaches a constant (when one exponent is zero and the other is negative).

3.3.2 Complex conjugate roots

The two solutions of (3.10) are

$$y_1(t) = e^{(\lambda+i\mu)t}, \quad y_2(t) = e^{(\lambda-i\mu)t},$$

where λ and μ are real with $\mu \neq 0$.

Recalling Euler's identity

$$e^{it} = \cos t + i \sin t \tag{3.14}$$

we have

$$y_1(t) = e^{\lambda t}e^{i\mu t} = e^{\lambda t}[\cos(\mu t) + i\sin(\mu t)]$$

and

$$y_2(t) = e^{\lambda t}e^{-i\mu t} = e^{\lambda t}[\cos(-\mu t) + i\sin(-\mu t)] = e^{\lambda t}[\cos(\mu t) - i\sin(\mu t)].$$

Hence we conclude that $y_1(t)$ and $y_2(t)$ are complex-valued solutions of (3.10).

Now note that, for example,

$$y_1(t) = e^{\lambda t}\cos(\mu t) + ie^{\lambda t}\sin(\mu t) = u + iv, \tag{3.15}$$

say, where u and v are real. Substituting into (3.10) we obtain

$$[au'' + bu' + cu] + i[av'' + bv' + cv] = 0.$$

However, when a complex number is zero both its real part and its imaginary part must be zero. Hence

$$au'' + bu' + cu = 0 \quad \text{and} \quad av'' + bv' + cv = 0,$$

that is, both $u = e^{\lambda t}\cos(\mu t)$ and $v = e^{\lambda t}\sin(\mu t)$ are real valued solutions of (3.10).

The Wronskian (see (3.4))

$$\begin{aligned} W[u,v](t) &= uv' - vu' \\ &= e^{\lambda t}\cos(\mu t)[e^{\lambda t}\sin(\mu t)]' - e^{\lambda t}\sin(\mu t)[e^{\lambda t}\cos(\mu t)]' \\ &= e^{\lambda t}\cos(\mu t)[e^{\lambda t}\mu\cos(\mu t) + \lambda e^{\lambda t}\sin(\mu t)] \\ &\quad -e^{\lambda t}\sin(\mu t)[-\mu e^{\lambda t}\sin(\mu t) + \lambda e^{\lambda t}\cos(\mu t)] \\ &= e^{2\lambda t}\mu, \end{aligned} \tag{3.16}$$

which is non-zero for all t and hence u and v are linearly independent.

By Theorem 3.7, the **general solution** is

$$\begin{aligned} y(t) &= c_1e^{\lambda t}\cos(\mu t) + c_2e^{\lambda t}\sin(\mu t), \\ &= e^{\lambda t}[c_1\cos(\mu t) + c_2\sin(\mu t)], \quad t \in \mathbb{R}. \end{aligned} \tag{3.17}$$

Example 3.16 Find the solution of the initial value problem

$$16y'' - 8y' + 145y = 0, \quad y(0) = -2, \quad y'(0) = 1\,.$$

Graph the solution and comment on its behaviour.

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3.3.3 Real, repeated roots

When $b^2 - 4ac = 0$ we have the repeated root

$$m_1 = m_2 = -\frac{b}{2a}\,.$$

Thus both roots yield the same solution

$$y_1(t) = e^{-\frac{bt}{2a}}$$

of (3.10). The general solution requires that we find a second linearly independent solution. We do this using a technique known as **reduction of order**.

3.3.4 Reduction of order[6]

[6] Reduction of order has been seen before in the MTH1002, part 2, lecture notes

The general idea is that, since $y_1(t)$ is a solution then so is $c\,y_1(t)$ for any constant c. However, this solution is not linearly independent.We generalise the method by replacing the constant c by a function $v(t)$ and we aim to determine $v(t)$ such that the product $v(t)y_1(t)$ is a linearly independent solution of (3.10).

We therefore postulate a solution

$$y = v(t)y_1(t) = v(t)e^{-\frac{bt}{2a}}$$

and substitute into (3.10), that is,

$$ay'' + by' + cy = 0$$

to determine $v(t)$. Using the product rule, we have

$$y' = v'e^{-\frac{bt}{2a}} - \frac{b}{2a}v\,e^{-\frac{bt}{2a}}$$

and

$$y'' = v''e^{-\frac{bt}{2a}} - \frac{b}{a}v'e^{-\frac{bt}{2a}} + \frac{b^2}{4a^2}v\,e^{-\frac{bt}{2a}}\,.$$

Now substitute into (3.10) to obtain

$$\left\{a\left[v'' - \frac{b}{a}v' + \frac{b^2}{4a^2}v\right] + b\left[v' - \frac{b}{2a}v\right] + cv\right\}e^{-\frac{bt}{2a}} = 0\,.$$

We cancel the (non-zero) exponential and rearrange the remaining terms to obtain

$$av'' + \left(\frac{b^2}{4a} - \frac{b^2}{2a} + c\right)v = 0\,.$$

However, we are considering the case of repeated roots, i.e. $b^2 = 4ac$. Hence the coefficient of v vanishes and we arrive at

$$v'' = 0\,.$$

Now set $w = v'$ to obtain

$$w' = 0\,,$$

i.e. we have reduced the order of the equation from second to first. The solution $w = c_1$ gives

$$v(t) = c_1 t + c_2$$

and so

$$y = c_1 te^{-\frac{bt}{2a}} + c_2 e^{-\frac{bt}{2a}}.$$

We have found that y is a linear combination of the two solutions

$$y_1(t) = e^{-\frac{bt}{2a}}, \quad y_2(t) = te^{-\frac{bt}{2a}}.$$

Their Wronskian is

$$\begin{aligned} W(t) &= \begin{vmatrix} e^{-\frac{bt}{2a}} & te^{-\frac{bt}{2a}} \\ -\frac{b}{2a}e^{-\frac{bt}{2a}} & \left(1 - \frac{bt}{2a}\right) e^{-\frac{bt}{2a}} \end{vmatrix} \\ &= e^{-\frac{bt}{a}}. \end{aligned}$$

Since $W[y_1, y_2](t)$ is never zero, the solutions y_1 and y_2 are a fundamental set of solutions and the **general solution** is

$$y = c_1 te^{-\frac{bt}{2a}} + c_2 e^{-\frac{bt}{2a}}, \quad t \in \mathbb{R},$$

i.e. one exponential solution corresponding to the repeated root and a second solution that is obtained by multiplying the exponential solution by t.

Example 3.17 Find the solution of the initial value problem

$$y'' - y' + \frac{y}{4} = 0, \quad y(0) = 2, \quad y'(0) = \frac{1}{3}.$$

Graph the solution and comment on its behaviour.

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Summary

We have shown that solutions of the second order, linear, homogeneous equation

$$ay'' + by' + cy = 0$$

are found by seeking solutions of the form $y = e^{mt}$. This leads to the characteristic polynomial

$$am^2 + bm + c = 0$$

and we must find its roots m_1 and m_2.

- If m_1 and m_2 are **real and not equal**, then the general solution is

$$\boxed{y = c_1 e^{m_1 t} + c_2 e^{m_2 t}}$$

- If m_1 and m_2 are **complex conjugates** $\lambda \pm i\mu$, then the general solution is

$$\boxed{y = c_1 e^{\lambda t} \cos(\mu t) + c_2 e^{\lambda t} \sin(\mu t)}$$

- If m_1 and m_2 are **real and equal**, then the general solution is

$$\boxed{y = c_1 e^{m_1 t} + c_2 t e^{m_1 t}}$$

3.4 Nonhomogeneous equations: Basic Theory

We now consider to the **nonhomogenous equation**

$$y'' + p(t)y' + q(t)y = g(t)\,, \tag{3.18}$$

where p, q and g are given continuous functions on the open interval $I : \alpha < t < \beta$. The equation

$$y'' + p(t)y' + q(t)y = 0\,, \tag{3.19}$$

with the same p and q but $g(t) \equiv 0$ is called the **corresponding homogeneous equation**.

Theorem 3.18 The general solution of the nonhomogeneous equation (3.18) can be written in the form

$$y(t) = c_1 y_1(t) + c_2 y_2(t) + y_p(t)\,, \tag{3.20}$$

where y_1 and y_2 are a fundamental set of solutions of the corresponding homogeneous equation (3.19), c_1 and c_2 are arbitrary constants, and y_p is any solution of the nonhomogeneous equation (3.18).

The method of solution for nonhomogeneous second order ODEs is then:

1. Find the general solution $c_1 y_1(t) + c_2 y_2(t)$ of the corresponding homogeneous equation. This is often called the **complementary solution** and is denoted by $y_c(t)$. We have already discussed how to find y_c when the homogeneous equation has constant coefficients.

2. Find a **particular solution** $y_p(t)$ of the nonhomogeneous equation. We will focus on how to do this below.

3. Add together y_c and y_p.

There are three methods for finding y_p that we discuss: the method of undetermined coefficients (§3.5), the method of variation of parameters (§3.6) and the method of reduction of order (§3.7).

3.5 Nonhomogeneous equations with constant coefficients[7]: Method of undetermined coefficients

[7] Nonhomogeneous ODEs with constant coefficients have been seen before in the MTH1002, part 2, lecture notes

Here we consider the nonhomogeneous equation with constant coefficients

$$ay'' + by' + cy = g(t)\,, \tag{3.21}$$

where a, b and c are constants and $a \neq 0$, and the corresponding homogeneous problem

$$ay'' + by' + cy = 0\,. \tag{3.22}$$

The method of undetermined coefficients requires that we make an 'intelligent guess' (a **trial solution**) about the form of the particular solution $y_p(t)$, but with the coefficients left unspecified. We then substitute the assumed form into the nonhomogeneous problem (3.21) and attempt to determine the coefficients such that y_p is a solution. If we cannot determine the coefficients then this means that our initial guess was incorrect and we must modify the assumption and try again. The method is used mainly when the homogeneous equation has constant coefficients (as in (3.22)) and the nonhomogeneous term is restricted to a small class of functions: polynomials, exponentials and sines and cosines.

3.5.1 Constructing the trial solution

- Find the general solution of the corresponding **homogeneous** problem, $y_c(t) = c_1 y_1(t) + c_2 y_2(t)$.
- Make sure that the nonhomogeneous term $g(t)$ involves nothing more than polynomials, exponentials and sines and cosines, or sums or products of such functions. If this is not the case, use the method of variation of parameters instead (see §3.6).
- Find a **particular solution of the nonhomogeneous problem** by assuming a trial solution of similar form to $g(t)$ (see the table below).
- The **general solution of the nonhomogeneous problem is** $y(t) = y_c(t) + y_p(t)$**.**

For IVPs: Use the initial conditions to determine the values of the arbitrary constants remaining in the general solution.

RHS $g(t)$	Trial solution $y_p(t)$
$A_k t^k + \cdots + A_1 t + A_0$	$(B_k t^k + \cdots + B_1 t + B_0)t^s$ where s is the number of times 0 is a root of the characteristic equation
$(A_k t^k + \cdots + A_1 t + A_0)e^{\alpha t}$	$(B_k t^k + \cdots + B_1 t + B_0)e^{\alpha t}t^s$ where s is the number of times α is a root of the characteristic equation
$(A_k t^k + \cdots + A_1 t + A_0)e^{\alpha t}\cos(\beta t)$ or $(A_k t^k + \cdots + A_1 t + A_0)e^{\alpha t}\sin(\beta t)$	$[(B_k t^k + \cdots + B_1 t + B_0)\cos(\beta t) + (C_k t^k + \cdots + C_1 t + C_0)\sin(\beta t)]e^{\alpha t}t^s$ where s is the number of times $\alpha + i\beta$ is a root of the characteristic equation

Example 3.19 Find the general solution of

$$y'' - 3y' - 4y = 3\exp(2t)\,.$$

Theorem 3.20 — Superposition principle for nonhomogeneous equations. A linear combination of nonhomogeneous terms $g_i(t)$, for $i = 1, \ldots, m$, implies a similar linear combination of particular integrals. That is, if $y_i(t)$, for $i = 1, \ldots, m$, are solutions of

$$y_i'' + p(t)y_i' + q(t)y_i = g_i(t)\,,$$

then

$$Y(t) = \sum_{i=1}^{m} c_i y_i(t)$$

is a solution of

$$Y'' + p(t)Y' + q(t)Y = G(t), \quad G(t) = \sum_{i=1}^{m} c_i g_i(t)\,,$$

for any set of constants c_i.

Activity 3.21 Prove Theorem 3.20.

3.6 Nonhomogeneous equations with variable coefficients: Variation of Parameters

Now we consider the nonhomogeneous problem with variable coefficients

$$y'' + p(t)y' + q(t)y = g(t)\,, \tag{3.24}$$

where p, q and g are given continuous functions on the open interval $I : \alpha < t < \beta$.

We begin by assuming[8] that we know the general solution

$$y_c(t) = c_1 y_1(t) + c_2 y_2(t) \tag{3.25}$$

of the corresponding homogeneous equation

$$y'' + p(t)y' + q(t)y = 0\,. \tag{3.26}$$

[8] Note that this is a major assumption because so far we have only considered solutions to the constant coefficient homogeneous equation. We meet a particular class of variable coefficient second order ODEs in §3.8.

Then, the idea is to replace the two constants c_1 and c_2 in (3.25) by two functions $u_1(t)$ and $u_2(t)$, say, so that we have

$$y = u_1(t)y_1(t) + u_2(t)y_2(t) \tag{3.27}$$

and we now aim to determine the unknown $u_1(t)$ and $u_2(t)$ in such a way that (3.27) is the solution of the nonhomogeneous problem (3.24).

We will do this by substituting (3.27) into (3.24), but since there are two unknowns and only one equation, we expect that there will be many possible choices of u_1 and u_2 that will suffice. Lagrange's method is to impose a (special[9]) second condition, giving two equations for two unknowns. We follow Lagrange's method.

[9] Lagrange showed how to choose this second condition in a special way to make the algebra much simpler than it otherwise would be.

First, we require the derivatives of y from (3.27) so that we can substitute them into (3.24). Thus, using the product rule, the first derivative is

$$y' = u_1(t)y_1'(t) + u_1'(t)y_1(t) + u_2(t)y_2'(t) + u_2'(t)y_2(t)\,.$$

Following Lagrange's method, we choose to make the terms involving u_1' and u_2' vanish, i.e. we set

$$u_1'(t)y_1(t) + u_2'(t)y_2(t) = 0 \tag{3.28}$$

and so the expression for y' becomes

$$y' = u_1(t)y_1'(t) + u_2(t)y_2'(t)\,. \tag{3.29}$$

Differentiating to find y'' we obtain

$$y'' = u_1(t)y_1''(t) + u_1'(t)y_1'(t) + u_2(t)y_2''(t) + u_2'(t)y_2'(t)\,. \tag{3.30}$$

Now we substitute y, y' and y'' into the nonhomogeneous equation (3.24). After gathering terms proportional to u_1 and u_2 we obtain

$$\begin{aligned} & u_1(t)\left[y_1''(t) + p(t)y_1'(t) + q(t)y_1(t)\right] \\ + \quad & u_2(t)\left[y_2''(t) + p(t)y_2'(t) + q(t)y_2(t)\right] \\ + \quad & u_1'(t)y_1'(t) + u_2'(t)y_2'(t) = g(t)\,. \end{aligned} \tag{3.31}$$

Now note that each of the terms in the square brackets is zero, because y_1 and y_2 satisfy the homogeneous problem (this is what we mean by (3.25)). Hence we are left with

$$u_1'(t)y_1'(t) + u_2'(t)y_2'(t) = g(t)\,. \tag{3.32}$$

Now we have two conditions, namely (3.28) and (3.32), that can be solved for the derivatives $u_1'(t)$ and $u_2'(t)$ of the unknown functions. We find

$$u_1'(t) = -\frac{y_2(t)g(t)}{W[y_1,y_2](t)}, \quad u_2'(t) = \frac{y_1(t)g(t)}{W[y_1,y_2](t)}\,,$$

where $W[y_1,y_2]$ is the Wronskian of y_1 and y_2 (note that since they are a fundamental set of solutions (see Theorem 3.7), their Wronskian is non-zero and we can divide by it). Finally we integrate to find

$$u_1(t) = -\int \frac{y_2(t)g(t)}{W[y_1,y_2](t)}\,dt + c_1, \quad u_2(t) = \int \frac{y_1(t)g(t)}{W[y_1,y_2](t)}dt + c_2\,.$$

Hence we have found the particular solution (3.27)

$$y_p(t) = -y_1(t)\int \frac{y_2(t)g(t)}{W[y_1,y_2](t)}\,dt + y_2(t)\int \frac{y_1(t)g(t)}{W[y_1,y_2](t)}dt$$

and, by Theorem 3.18, the general solution is

$$y = c_1y_1(t) + c_2y_2(t) + y_p(t)\,.$$

We summarise the result as a theorem.

Theorem 3.22 Consider the nonhomogeneous equation

$$y'' + p(t)y' + q(t)y = g(t)\,, \tag{3.33}$$

where p, q and g are given continuous functions on the open interval $I : \alpha < t < \beta$.

If the functions $y_1(t)$ and $y_2(t)$ are linearly independent solutions of the corresponding homogeneous equation

$$y'' + p(t)y' + q(t)y = 0\,, \tag{3.34}$$

then a particular solution of (3.33) is

$$y_p(t) = -y_1(t)\int_{t_0}^{t} \frac{y_2(s)g(s)}{W[y_1,y_2](s)}\,ds + y_2(t)\int_{t_0}^{t} \frac{y_1(s)g(s)}{W[y_1,y_2](s)}\,ds\,, \tag{3.35}$$

where $t_0 \in I$, and the general solution of (3.33) is

$$y = c_1y_1(t) + c_2y_2(t) + y_p(t)\,, \tag{3.36}$$

where c_1 and c_2 are constants.

Example 3.23 Use the method of variation of parameters to find the general solution of

$$y'' + 4y' + 4y = t^{-2}e^{-2t}, \quad t > 0\,.$$

3.7 Nonhomogeneous equations with variable coefficients: Reduction of Order

This is a powerful method to find the solution of a nonhomogeneous equation with variable coefficients given that one solution of the corresponding homogeneous problem is already known. We met a simpler version of it before in §3.3.4.

Suppose that we know one non-trivial solution $y_1(t)$ of the homogeneous equation

$$y'' + p(t)y' + q(t)y = 0 \tag{3.37}$$

and we want to find the solution of the nonhomogeneous equation

$$y'' + p(t)y' + q(t)y = g(t)\,. \tag{3.38}$$

We set

$$y(t) = v(t)y_1(t)$$

so that,

$$y' = vy_1' + v'y_1, \quad y'' = vy_1'' + 2v'y_1' + v''y_1\,.$$

Now substitute these derivatives into the nonhomogeneous equation (3.38) to obtain, after some rearranging

$$v[y_1'' + p(t)y_1' + q(t)y_1(t)] + v''y_1 + v'[2y_1' + p(t)y_1] = g(t)\,. \tag{3.39}$$

However, y_1 is a solution of the homogeneous equation (3.37) and so the first square bracket vanishes and we obtain

$$w'y_1 + w\,[2y_1' + p(t)y_1] = g(t)\,, \tag{3.40}$$

where we have set $w = v'$. This is now a first order equation for w which can be solved using an integrating factor.

Example 3.24 Solve

$$y'' - \frac{(4t+1)}{t}y' + \frac{2}{t}(2t+1)y = 4e^{2t}$$

on $t \in (0, \infty)$, given that one solution of the corresponding homogeneous equation is $y = e^{2t}$.

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3.8 Euler-Cauchy equations

These are a special class of homogeneous equation with specific variable coefficients. Euler-Cauchy equations take the form

$$t^2y'' + \alpha ty' + \beta y = 0, \quad \text{for } t > 0, \tag{3.41}$$

where α, β are constants.

We seek trial solutions of the form

$$y = t^r\,,$$

where r is a constant that is to be determined. Differentiating gives

$$y' = rt^{r-1}, \quad y'' = r(r-1)t^{r-2}\,.$$

Substituting into (3.41) yields

$$t^r\left[r(r-1) + \alpha r + \beta\right] = 0$$

and so we are required to solve the **auxiliary equation** (or **indicial equation**)

$$r(r-1) + \alpha r + \beta = 0\,. \tag{3.42}$$

The roots are

$$r_1, r_2 = -\frac{\alpha - 1}{2} \pm \frac{\sqrt{(\alpha-1)^2 - 4\beta}}{2}\,. \tag{3.43}$$

As we did for second order linear equations with constant coefficients, it is necessary to consider separately the cases in which the roots are real and different, real and equal, and complex conjugates.

3.8.1 Real and different roots

If $r_1 \neq r_2$ then we obtain the two solutions

$$y_1 = t^{r_1}, \quad y_2 = t^{r_2}.$$

Since

$$W[y_1, y_2] = (r_2 - r_1)t^{r_1+r_2-1},$$

which is non-zero for $t > 0$ and $r_1 \neq r_2$, it follows that the general solution is

$$y = c_1 t^{r_1} + c_2 t^{r_2}, \quad t > 0. \tag{3.44}$$

Note that if r is not a rational number, then t^r is defined by $t^r = e^{r \ln t}$.

Example 3.25 Solve

$$2t^2y'' + 3ty' - y = 0, \quad t > 0.$$

3.8.2 Real and equal roots

Here $r_1 = r_2 = -\frac{\alpha-1}{2} = r$, say, and we have only one solution

$$y_1 = t^r\,.$$

We then use the method of reduction of order (see §3.7) to determine the second solution. Writing

$$y = t^r v(t),$$

finding y' and y'' and substituting into the Euler-Cauchy equation (3.41) leads to (see activity 3.26)

$$t^2 v'' + t(2r+\alpha)v' = 0\,. \tag{3.45}$$

Equation (3.45) can be solved by separation of variables. We rewrite the equation as

$$\frac{v''}{v'} = -\frac{2r+\alpha}{t}\,,$$

since $t \neq 0$ and we assume that $v' \neq 0$. Integrating gives

$$\ln|v'| = -(2r+\alpha)\ln|t| + \ln c$$

where $\ln c$ is a constant. Hence

$$\begin{aligned} |v'| &= e^{-(2r+\alpha)\ln|t|+\ln c} = e^{\ln|t|^{-(2r+\alpha)}} e^{\ln c} \\ \Rightarrow v' &= d\,t^{-(2r+\alpha)}, \quad d = \pm c, \quad t > 0, \end{aligned} \tag{3.46}$$

using laws of logarithms and exponents.

Now, using the definition $r = r_1 = r_2$ we have

$$2r + \alpha = -2\frac{(\alpha-1)}{2} + \alpha = -1$$

and hence

$$v' = dt^{-1},$$

and integrating gives

$$v = d\ln t + f\,,$$

where f is a constant. Hence

$$y = t^r(d\ln t + f) = c_2 y_2 + c_1 y_1, \quad t > 0 \tag{3.47}$$

is the general solution, as it can be shown (see activity 3.27) that $W[y_1, y_2] = t^{2r-1}$ and y_1 and y_2 are linearly independent for $t > 0$.

Activity 3.26 Derive the ODE (3.45) for $v(t)$ above using the method of reduction of order.

Activity 3.27 Show that $y_1 = t^r$ and $y_2 = t^r \ln t$ are linearly independent for $t > 0$.

Example 3.28 Solve

$$t^2y'' + 5ty + 4y = 0, \quad t > 0\,.$$

3.8.3 Complex roots

Here the roots are complex conjugates, $r_1 = \lambda + i\mu$ and $r_2 = \lambda - i\mu$, say, with $\mu \neq 0$. First of all we need to define what we mean by t^r when r is complex. Recall that

$$t^r = e^{r \ln t}$$

when $t > 0$ and r is real. We use this same equation when r is complex. Thus

$$\begin{aligned} t^{\lambda+i\mu} &= e^{(\lambda+i\mu)\ln t} = e^{\lambda \ln t}\, e^{i\mu \ln t} = t^\lambda\, e^{i\mu \ln t} \\ &= t^\lambda \left[\cos(\mu \ln t) + i \sin(\mu \ln t)\right], \quad t > 0\,. \end{aligned} \tag{3.48}$$

The general solution is

$$y = c_1 t^{\lambda+i\mu} + c_2 t^{\lambda-i\mu}\,.$$

Note however that this is a complex-valued function. As was the case for the second order differential equation with constant coefficients with complex roots, it can be shown that the real and imaginary parts of $t^{\lambda+i\mu}$, namely

$$t^\lambda \cos(\mu \ln t) \quad \text{and} \quad t^\lambda \sin(\mu \ln t)\,,$$

are also solutions of the Euler-Cauchy equation (3.41) and their Wronskian

$$W[t^\lambda \cos(\mu \ln t), t^\lambda \sin(\mu \ln t)] = \mu t^{2\lambda-1}$$

is non-zero for $t > 0$ and $\mu \neq 0$. Hence the general solution for $t > 0$ can be written

$$y = c_1 t^\lambda \cos(\mu \ln t) + c_2 t^\lambda \sin(\mu \ln t), \quad t > 0\,. \tag{3.49}$$

Example 3.29 Solve

$$t^2y'' + ty' + y = 0, \quad t > 0.$$

Activity 3.30 As an alternative way of finding solutions to the Euler-Cauchy equation (3.41), use the change of variables $t = e^s$ and show that $y(t) = y(e^s) = Y(s)$ satisfies a linear ODE with constant coefficients. Hence find $y(t)$.

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3.9 Method of undetermined coefficients for nonhomogeneous Euler-Cauchy equations

A modified method of undetermined coefficients can be used to solve a nonhomogeneous Euler-Cauchy equation of the form

$$t^2y'' + \alpha ty' + \beta y = \gamma t^\delta$$

by considering trial solutions of the form $y = ct^\delta$ (for the non-resonant case). If that fails, try $y = ct^\delta \ln t$ or $y = ct^\delta (\ln t)^2$ (for the simple resonance and double resonance cases, respectively).

4 Systems of First Order Linear Ordinary Differential Equations

Reference: Boyce & DiPrima, Chapter 6

Systems of ordinary differential equations arise when there is a single independent variable, t, say, and n dependent variables $x_1(t), x_2(t), \ldots x_n(t)$ (the unknowns), each of which is a function of t. Physically, we are are modelling problems in which there are a number of separate elements that are linked together in some way. Mathematically, this structure is very useful because any higher-order ODE can be reformulated as a system of first-order ODEs.

4.1 General framework

The most general system of n ordinary differential equations for $x_1(t), x_2(t), \ldots x_n(t)$ has the form

$$\begin{aligned} \frac{dx_1}{dt} &= f_1(x_1, x_2, \ldots, x_n)\,, \\ \frac{dx_2}{dt} &= f_2(x_1, x_2, \ldots, x_n)\,, \\ &\vdots \\ \frac{dx_n}{dt} &= f_n(x_1, x_2, \ldots, x_n)\,. \end{aligned} \tag{4.1}$$

The system (4.1) has a **solution** on the interval $I : \alpha < t < \beta$ if there exists a set of n functions

$$x_1(t), \quad x_2(t), \quad \ldots, \quad x_n(t) \tag{4.2}$$

that are differentiable at all points in the interval I and that satisfy the equations (4.1) at all points in this interval. We might also be given **initial conditions**

$$x_1(t_0) = x_1^0, \quad x_2(t_0) = x_2^0, \quad \ldots \quad x_n(t_0) = x_n^0\,, \tag{4.3}$$

where t_0 is a specified value of t in the interval I and the $x_1^0, x_2^0, \ldots, x_n^0$ are given numbers. The differential equations (4.1) and the initial conditions (4.3) together form an **initial value problem**.

The solution (4.2) can be viewed as a set of parametric equations in an n-dimensional space: for a given value of the parameter t, equations (4.2) tell us the coordinates $x_1, x_2, \ldots, x_n$ of a point in space. As t changes, the coordinates change and the collection of points for $\alpha < t < \beta$ form a curve. We can think of the curve as the path (or trajectory) of a particle moving according to the system of equations (4.1). The initial conditions (4.3) determine the starting point of the moving particle.

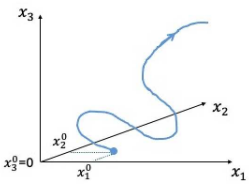


Figure 4.1: A particle that at $t = t_0$ is located at position (x_1^0, x_2^0, x_3^0) moves along the path described by the parametric curve (4.2).

In vector notation, the system of first order ODEs (4.1) can be written as

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{f}(\boldsymbol{x}, t)\,, \tag{4.4}$$

where

$$\boldsymbol{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}, \quad \boldsymbol{f}(\boldsymbol{x}, t) = \begin{pmatrix} f_1(\boldsymbol{x}, t) \\ f_2(\boldsymbol{x}, t) \\ \vdots \\ f_n(\boldsymbol{x}, t) \end{pmatrix}, \tag{4.5}$$

i.e. $\boldsymbol{x}$ is a column vector (or $n \times 1$ matrix) whose elements are the unknown functions of t, and $\boldsymbol{f}$ is a column vector of the right-hand sides of (4.1). The initial conditions (4.3) are written as

$$\boldsymbol{x}(t_0) = \boldsymbol{k}\,. \tag{4.6}$$

It is apparent that we need to deal with **matrix functions**, i.e. matrices whose elements are functions of t.

Example 4.1 Consider a particle that moves on a curve defined by

$$\boldsymbol{x}(t) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}.$$

Find $\frac{d\boldsymbol{x}}{dt}$ and interpret your result geometrically.

.

4.2 Revision: Matrices and linear algebraic systems

In order to be able to deal with systems of differential equations, students are strongly encouraged to revise the material in MTH1002 on matrices and linear algebraic systems. In particular, students should be comfortable with the material summarised in Appendices A and B (see handouts).

4.3 Basic theory of systems of first order linear ODEs

Definition 4.2 If each of the functions $f_1, \ldots, f_n$ in system (4.1) is a linear function of the dependent variables $x_1, \ldots, x_n$, then the system of equations is said to be **linear**. Otherwise it is **nonlinear**.

The most general form of n first order linear ordinary differential equations is

$$\begin{aligned} \frac{dx_1}{dt} &= p_{11}(t)x_1 + \cdots + p_{1n}(t)x_n + g_1(t)\,, \\ \frac{dx_2}{dt} &= p_{21}(t)x_1 + \cdots + p_{2n}(t)x_n + g_2(t)\,, \\ &\vdots \\ \frac{dx_n}{dt} &= p_{n1}(t)x_1 + \cdots + p_{nn}(t)x_n + g_n(t)\,. \end{aligned} \tag{4.7}$$

We write the system (4.7) in matrix form as

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} + \boldsymbol{g}(t)\,. \tag{4.8}$$

The $n \times n$ matrix

$$\boldsymbol{P} = \begin{pmatrix} p_{11}(t) & p_{12}(t) & \ldots & p_{1n}(t) \\ p_{21}(t) & p_{22}(t) & \ldots & p_{2n}(t) \\ \vdots & \vdots & \ldots & \vdots \\ p_{n1}(t) & p_{n2}(t) & \ldots & p_{nn}(t) \end{pmatrix}$$

is called the matrix of coefficients and the column vector

$$\boldsymbol{g}(t) = \begin{pmatrix} g_1(t) \\ g_2(t) \\ \vdots \\ g_n(t) \end{pmatrix}$$

is the vector of free terms. The solution is

$$\boldsymbol{x}(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix} \tag{4.9}$$

and initial conditions (4.3) or (4.6) are

$$\boldsymbol{x}(t_0) = \boldsymbol{k}\,. \tag{4.10}$$

Theorem 4.3 If $\boldsymbol{P}$ and $\boldsymbol{g}$ are continuous on an open interval $I : \alpha < t < \beta$ (i.e. the functions $p_{11}, p_{12}, \ldots, p_{nn}, g_1, \ldots, g_n$ are continuous on I), with $t_0 \in I$, then there exists a **unique solution** of the linear system (4.8) that satisfies the initial conditions (4.10). The solution exists throughout the interval I.

Definition 4.4 A **homogeneous** system of linear ODEs is a linear system in which $\boldsymbol{g}(t) = \mathbf{0}$ for all t in the interval I. Hence (4.8) becomes

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x}. \tag{4.11}$$

As before, we'll see that once the homogeneous system has been solved there are methods to solve the **nonhomogeneous** ($\boldsymbol{g} \neq \mathbf{0}$) system (see §4.7).

Theorem 4.5 — Principle of superposition for homogeneous linear systems. If the vector functions $\boldsymbol{x}^{(1)}(t)$ and $\boldsymbol{x}^{(2)}(t)$ are solutions of the homogeneous system (4.11) then the linear combination

$$\boldsymbol{x}(t) = c_1\boldsymbol{x}^{(1)}(t) + c_2\boldsymbol{x}^{(2)}(t)$$

is also a solution for any constants c_1 and c_2.

We'll use the notation

$$\boldsymbol{x}^{(1)}(t) = \begin{pmatrix} x_{11}(t) \\ x_{21}(t) \\ \vdots \\ x_{n1}(t) \end{pmatrix}, \quad \ldots, \quad \boldsymbol{x}^{(k)}(t) = \begin{pmatrix} x_{1k}(t) \\ x_{2k}(t) \\ \vdots \\ x_{nk}(t) \end{pmatrix}, \ldots \tag{4.12}$$

to denote the specific solutions of (4.11), where $x_{ij}(t) = x_i^{(j)}(t)$ refers to the ith component of the jth solution $\boldsymbol{x}^{(j)}(t)$.

We see that $\boldsymbol{x}(t) = \mathbf{0}$ is always a solution to (4.11). We call it the **trivial solution**.

Example 4.6 Write the linear system

$$\frac{dx_1}{dt} = x_1(t) + x_2(t),$$
$$\frac{dx_2}{dt} = 4x_1(t) + x_2(t),$$

in the matrix form (4.11). Then verify that

$$\boldsymbol{x}^{(1)}(t) = \begin{pmatrix} e^{3t} \\ 2e^{3t} \end{pmatrix}, \quad \boldsymbol{x}^{(2)}(t) = \begin{pmatrix} e^{-t} \\ -2e^{-t} \end{pmatrix}$$

are solutions of (4.11). Plot the vector function $\boldsymbol{x}^{(1)}(t)$.

.

Activity 4.7 According to Theorem 4.5

$$\boldsymbol{x} = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-t}$$

also satisfies (4.11), for arbitrary constants c_1 and c_2. Verify this.

.

Definition 4.8 The determinant

$$W[\boldsymbol{x}^{(1)},\ldots,\boldsymbol{x}^{(n)}] = \left|\boldsymbol{x}^{(1)}\,\boldsymbol{x}^{(2)}\,\ldots\,\boldsymbol{x}^{(n)}\right| = \begin{vmatrix} x_{11}(t) & \ldots & x_{1n}(t) \\ \vdots & & \vdots \\ x_{n1}(t) & \ldots & x_{nn}(t) \end{vmatrix} \tag{4.13}$$

is called the **Wronskian of the n solutions** $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(n)}$ of system (4.11).

Theorem 4.9 If $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(n)}$ are solutions of system (4.11) on the interval $I : \alpha < t < \beta$, then in this interval the Wronskian $W[\boldsymbol{x}^{(1)},\ldots,\boldsymbol{x}^{(n)}]$ either is identically zero or it never vanishes. The solutions are **linearly independent** on I if and only if

$$W[\boldsymbol{x}^{(1)},\ldots,\boldsymbol{x}^{(n)}] \neq 0\,.$$

Theorem 4.10 If $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(n)}$ are linearly independent solutions of system (4.11) on the interval $I : \alpha < t < \beta$, then each solution $\boldsymbol{x}(t)$ of that system can be expressed as a linear combination

$$\boldsymbol{x}(t) = c_1\boldsymbol{x}^{(1)}(t) + \cdots + c_n\boldsymbol{x}^{(n)}(t) \tag{4.14}$$

in exactly one way. We call (4.14) the **general solution** of (4.11). The set of solutions $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(n)}$ is said to be a **fundamental set of solutions** for the interval I.

4.4 Homogeneous linear systems with constant coefficients

We consider the simplest case

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{A}\boldsymbol{x}\,, \tag{4.15}$$

where $\boldsymbol{A}$ is a constant $n \times n$ matrix. This is sometimes called an **autonomous ODE system** because $\boldsymbol{A}$ is not a function of the independent variable t.

If $n = 1$ we have a single first order differential equation

$$\frac{dx}{dt} = ax, \quad a \neq 0 \tag{4.16}$$

with solution $x = ce^{at}$. The **equilibrium solution**[11] is defined by $dx/dt = 0$ and hence the only equilibrium solution is $x = 0$. Figures 4.2 and 4.3 show the solutions in the xt-plane (the integral curves). If $a > 0$, solutions move away from $x = 0$. If $a < 0$, solutions move towards the equilibrium solution.

[11] aka equilibrium point, critical point or fixed point

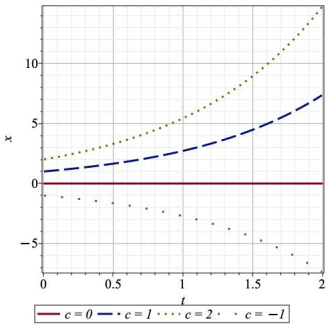


Figure 4.2: The solutions to (4.16), namely $x(t) = ce^{at}$ for different c with $a = 1 > 0$, move away from the equilibrium $x = 0$.

For higher order systems we follow a similar method: Equilibrium solutions of (4.15) are found by solving

$$A\boldsymbol{x} = \boldsymbol{0}\,.$$

If $|\boldsymbol{A}| \neq 0$ then $\boldsymbol{x} = \boldsymbol{0}$ is the only equilibrium solution. It is of interest whether other solutions approach the equilibrium or move away from it as t increases.

Visualisation is easiest if we take the case $n = 2$. We let $\boldsymbol{x} = \boldsymbol{x}(t)$ be a solution of (4.15) for $t \in I$, with the initial condition $\boldsymbol{x}(t_0) \neq \boldsymbol{0}$, $t_0 \in I$. Then, as t varies, $\boldsymbol{x}(t) = (x_1(t), x_2(t))$ describes a parametric curve in the x_1x_2-plane with t as the parameter. The x_1x_2-plane is referred to as the **phase space** (or phase plane for the two-dimensional case). The **phase portrait** of system (4.15) is a phase-space plot of a representative set of solution curves (sometimes called trajectories) $\boldsymbol{x}(t)$ corresponding to different $\boldsymbol{x}(t_0)$. An example of a phase portrait is shown in figure 4.5.

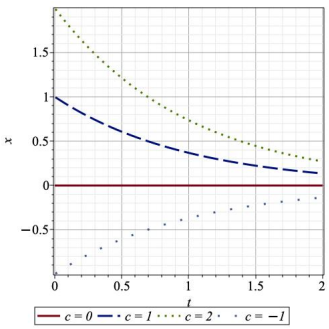


Figure 4.3: As for figure 4.2 except $a = -1 < 0$, showing that solutions approach the equilibrium.

To find the general solution of (4.15) we proceed in a similar way as we did for second order linear ODEs with constant coefficients, i.e. we seek solutions in the form of an exponential

$$\boldsymbol{x}(t) = \boldsymbol{\xi}e^{rt}\,, \tag{4.17}$$

where the (constant) exponent r and the (constant) vector $\boldsymbol{\xi}$ are to be determined. Substituting (4.17) into (4.15) gives

$$r\boldsymbol{\xi}e^{rt} = \boldsymbol{A}\boldsymbol{\xi}e^{rt}\,.$$

Hence we require that

$$\boldsymbol{A}\boldsymbol{\xi} = r\boldsymbol{\xi}\,, \tag{4.18}$$

i.e. the vector $\boldsymbol{x}$ given by (4.17) is a solution of the system (4.15) provided that r is an eigenvalue and $\boldsymbol{\xi}$ is the corresponding eigenvector of the matrix $\boldsymbol{A}$.

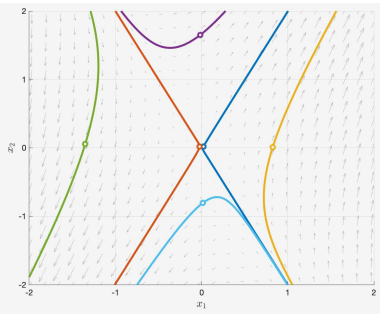


Figure 4.4: An example of a phase portrait.

To find the eigenvalues we must find the roots of the characteristic equation

$$|\boldsymbol{A} - r\boldsymbol{I}| = 0\,,$$

which is a polynomial of degree n. Similar to the case for second order ODEs, when $n = 2$ the roots can be either:

- Real and different
- Complex conjugates
- Real and repeated

We know by Theorem 4.10 that the general solution of an $n \times n$ linear system is a linear combination of n linearly independent solutions. Hence, if the matrix $\boldsymbol{A}$ has n linearly independent eigenvectors, $\boldsymbol{\xi}^{(j)}$ for $j = 1, \ldots, n$, with corresponding eigenvalues r_j, then the general solution of the homogeneous system (4.15) is

$$\boldsymbol{x}(t) = c_1\boldsymbol{\xi}^{(1)}e^{r_1 t} + \cdots + c_n\boldsymbol{\xi}^{(n)}e^{r_n t}\,. \tag{4.19}$$

4.4.1 Real and different eigenvalues

Example 4.11 Find the general solution of the autonomous system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix}\boldsymbol{x}$$

and sketch the phase portrait.

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Activity 4.12 Find the general solution of the autonomous system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} -3 & \sqrt{2} \\ \sqrt{2} & -2 \end{pmatrix} \boldsymbol{x}$$

and sketch the phase portrait.

.

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4.4.2 Complex conjugate eigenvalues

For a real matrix $\boldsymbol{A}$, complex eigenvalues appear in complex conjugate pairs. For example, if $r_1 = \lambda + i\mu$ (where λ and μ are real) is an eigenvalue then so is $r_2 = \lambda - i\mu$. Furthermore, the corresponding eigenvectors $\boldsymbol{\xi}^{(1)}$ and $\boldsymbol{\xi}^{(2)}$ are also complex conjugates. This can be shown by considering the equation for r_1 and $\boldsymbol{\xi}^{(1)}$

$$(\boldsymbol{A} - r_1\boldsymbol{I})\boldsymbol{\xi}^{(1)} = \boldsymbol{0} \tag{4.20}$$

and taking the complex conjugate to obtain

$$(\boldsymbol{A} - \bar{r_1}\boldsymbol{I})\bar{\boldsymbol{\xi}}^{(1)} = \boldsymbol{0}\,, \tag{4.21}$$

since $\boldsymbol{A}$ and $\boldsymbol{I}$ are real. Hence $r_2 = \bar{r_1}$ is an eigenvalue with eigenvector $\boldsymbol{\xi}^{(2)} = \bar{\boldsymbol{\xi}}^{(1)}$. Thus

$$\boldsymbol{x}^{(1)}(t) = \boldsymbol{\xi}^{(1)}e^{r_1 t}, \quad \boldsymbol{x}^{(2)}(t) = \bar{\boldsymbol{\xi}}^{(1)}e^{\bar{r_1}t}, \tag{4.22}$$

are solutions of (4.15) and are complex conjugates of each other.

As before, we can find two real-valued solutions corresponding to the eigenvalues r_1 and r_2 by taking the real and imaginary parts of $\boldsymbol{x}^{(1)}$ (or $\boldsymbol{x}^{(2)}$). If we write $\boldsymbol{\xi}^{(1)} = \boldsymbol{a} + i\boldsymbol{b}$, where $\boldsymbol{a}$ and $\boldsymbol{b}$ are real, then we have

$$\begin{aligned}
\boldsymbol{x}^{(1)}(t) &= (\boldsymbol{a} + i\boldsymbol{b})e^{(\lambda+i\mu)t} \\
&= (\boldsymbol{a} + i\boldsymbol{b})e^{\lambda t}(\cos\mu t + i\sin\mu t) \\
&= e^{\lambda t}(\boldsymbol{a}\cos\mu t - \boldsymbol{b}\sin\mu t) + ie^{\lambda t}(\boldsymbol{a}\sin\mu t + \boldsymbol{b}\cos\mu t) \\
&= \boldsymbol{m}(t) + i\boldsymbol{n}(t)\,,
\end{aligned} \tag{4.23}$$

say. It can be shown that $\boldsymbol{m}$ and $\boldsymbol{n}$ are linearly independent solutions.

If a real $n \times n$ matrix $\boldsymbol{A}$ has two complex conjugate eigenvalues $r_1 = \lambda + i\mu$ and $r_2 = \lambda - i\mu$ with corresponding eigenvectors $\boldsymbol{\xi}^{(1)} = \boldsymbol{a} + i\boldsymbol{b}$ and $\boldsymbol{\xi}^{(2)} = \boldsymbol{a} - i\boldsymbol{b}$ and the other eigenvalues $r_3, \ldots, r_n$ are real and distinct with eigenvectors $\boldsymbol{\xi}^{(3)}, \ldots, \boldsymbol{\xi}^{(n)}$, then the general solution of (4.15) is

$$\boldsymbol{x} = c_1\boldsymbol{m}(t) + c_2\boldsymbol{n}(t) + c_3\boldsymbol{\xi}^{(3)}e^{r_3 t} + \cdots + c_n\boldsymbol{\xi}^{(n)}e^{r_n t}, \tag{4.24}$$

where $\boldsymbol{m}$ and $\boldsymbol{n}$ are given by (4.23).

Example 4.13 Find a fundamental set of real-valued solutions of the system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} -\frac{1}{2} & 1 \\ -1 & -\frac{1}{2} \end{pmatrix} \boldsymbol{x}$$

.

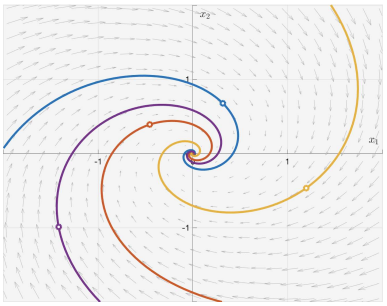


Figure 4.5: Phase portrait for example 4.13

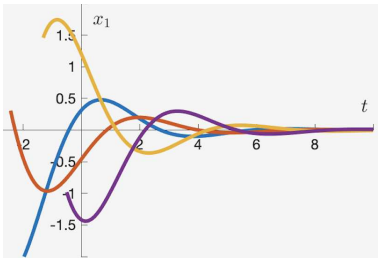


Figure 4.6: Plots of $x_1(t)$ for example 4.13.

4.4.3 Repeated eigenvalues

If $r = \rho$ is a k-fold root of the characteristic equation then ρ is said to be an eigenvalue of multiplicity k of the matrix $\boldsymbol{A}$. There are then two possibilities:

1. There are k linearly independent eigenvectors $\boldsymbol{\xi}^{(1)}, \ldots, \boldsymbol{\xi}^{(k)}$ corresponding to the eigenvalue ρ. In this case the general solution is
$$\boldsymbol{x} = c_1 e^{\rho t}\boldsymbol{\xi}^{(1)} + \cdots + c_k e^{\rho t}\boldsymbol{\xi}^{(k)},$$
as expected.

2. There are fewer than k such eigenvectors. In this case it is necessary to find other solutions of a different form. After finding solutions of the form
$$\boldsymbol{x} = \boldsymbol{\xi} e^{rt}$$
we then, following the procedure set out for second order linear equations, try solutions of the form
$$\boldsymbol{x} = \boldsymbol{\xi} t e^{rt} + \boldsymbol{\eta} e^{rt}$$
and then
$$\boldsymbol{x} = \boldsymbol{\theta} t^2 e^{rt} + \boldsymbol{\xi} t e^{rt} + \boldsymbol{\eta} e^{rt}$$
etc., until we find a complete set of linearly independent solutions.

Example 4.14 Find the general solution of the autonomous system
$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix} \boldsymbol{x} \tag{4.25}$$
and draw a phase portrait for this system.

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4.5 Stability

Reference: Boyce & DiPrima, Chapter 9 (but note some change in notation concerning the ordering of the eigenvalues)

We seek a qualitative understanding of the behaviour of solutions of the homogeneous system of linear, first order ODEs

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{A}\boldsymbol{x} \tag{4.26}$$

using geometrical methods. We concentrate on the case in which $\boldsymbol{A}$ is a 2×2 non-singular matrix (i.e. $|\boldsymbol{A}| \neq 0$) so that $\boldsymbol{x} = \boldsymbol{0}$ is the only **equilibrium solution** satisfying $d\boldsymbol{x}/dt = \boldsymbol{0}$.

Recall that in phase space[12], generic (non-equilibrium) solutions $\boldsymbol{x} = \boldsymbol{x}(t)$ are parametric curves that show the trajectory of a moving particle whose velocity $d\boldsymbol{x}/dt = \boldsymbol{A}\boldsymbol{x}$. Equilibrium solutions do not move and are often called **equilibrium points** (or fixed points, or critical points). A representative set of trajectories and equilibrium points in the phase space is called a **phase portrait**. Our aim is to characterise the differential equation according to the geometrical shape of its phase portrait. Several different cases need to be considered depending on the form of the eigenvalues of $\boldsymbol{A}$, r_1 and r_2.

[12] i.e. the x_1x_2-plane

4.5.1 Real and different eigenvalues with the same sign

The general solution of (4.26) is

$$\boldsymbol{x} = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = c_1\boldsymbol{\xi}^{(1)}e^{r_1 t} + c_2\boldsymbol{\xi}^{(2)}e^{r_2 t}, \tag{4.27}$$

where r_1 and r_2 are either both positive or both negative.

Consider first the case in which the eigenvalues are both negative with $r_1 < r_2 < 0$ and that the eigenvectors are as shown in figure 4.7. Then, (4.27) states that as $t \to \infty$, $\boldsymbol{x} \to \boldsymbol{0}$ regardless of the values of c_1 and c_2. Hence all solutions approach the equilibrium point at long times. If the solution starts at a point on the line through $\boldsymbol{\xi}^{(1)}$ then we know that $c_2 = 0$ and the solution remains on that line for all t, approaching the origin as $t \to \infty$. Similarly, if the initial conditions are such that a solution starts on the line through $\boldsymbol{\xi}^{(2)}$ then $c_1 = 0$ and solutions approach the origin along $\boldsymbol{\xi}^{(2)}$. More generally, it is helpful to write the solution (4.27) in the form

$$\boldsymbol{x} = e^{r_2 t}\left[c_1\boldsymbol{\xi}^{(1)}e^{(r_1 - r_2)t} + c_2\boldsymbol{\xi}^{(2)}\right]. \tag{4.28}$$

Then, since $r_1 - r_2 < 0$, the second term $c_2\boldsymbol{\xi}^{(2)}$ dominates the first for sufficiently large t, provided that $c_2 \neq 0$. Hence, as $t \to \infty$, the trajectory approaches the origin tangent to $\boldsymbol{\xi}^{(2)}$. Figure 4.7 shows an example phase portrait. Figure 4.8 shows examples of some solutions $x_1(t)$ ($x_2(t)$ are similar). This type of equilibrium point is called a **stable node** (aka nodal sink).

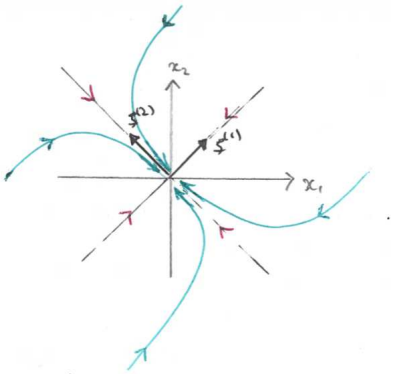


Figure 4.7: The phase portrait of a stable node with $r_1 < r_2 < 0$.

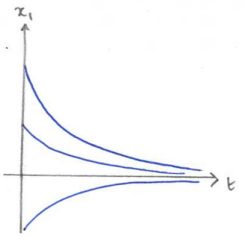


Figure 4.8: Examples of the solutions $x_1(t)$ for different initial conditions for the same case as Figure 4.7.

Now consider the case when the two eigenvalues are positive with $0 < r_1 < r_2$. In this case both $e^{r_1 t}$ and $e^{r_2 t}$ grow and so solutions move away from the origin. Since $r_1 - r_2 < 0$, the second term of (4.28) $c_2\boldsymbol{\xi}^{(2)}$ dominates the first for sufficiently large t, provided that $c_2 \neq 0$. Hence, as $t \to \infty$, the trajectory moves away from the origin parallel to $\boldsymbol{\xi}^{(2)}$. This type of equilibrium point is called a **unstable node** (aka nodal source).

From our previous examples, Example 4.12 is a stable node.

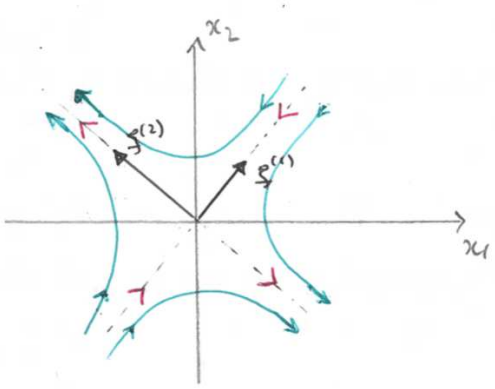


Figure 4.9: The phase portrait of a saddle point with $r_1 < 0 < r_2$.

4.5.2 Real and different eigenvalues with opposite sign

Now consider the case $r_1 < 0 < r_2$ and the solution

$$\boldsymbol{x} = c_1\boldsymbol{\xi}^{(1)}e^{r_1 t} + c_2\boldsymbol{\xi}^{(2)}e^{r_2 t}\,. \tag{4.29}$$

If the initial condition is on the line through $\boldsymbol{\xi}^{(1)}$ then $c_2 = 0$ and the solution remains on $\boldsymbol{\xi}^{(1)}$ for all time. Since $r_1 < 0$, $|\boldsymbol{x}| \to 0$ and the particle approaches the origin. Conversely, if the particle starts on the line through $\boldsymbol{\xi}^{(2)}$ then $c_1 = 0$ and the solution remains on $\boldsymbol{\xi}^{(2)}$ for all time. However, since $r_2 > 0$, $|\boldsymbol{x}| \to \infty$ and the solution moves away from the origin. For more generic initial conditions, eventually the solutions approach infinity parallel to the eigenvector $\boldsymbol{\xi}^{(2)}$ because the positive exponential term dominates for large t. An example of a phase portrait is shown in Figure 4.9.

Figure 4.10 shows examples of some solutions $x_1(t)$. For certain special initial conditions the positive exponential term is absent and $x_1 \to 0$ as $t \to 0$, while for all other initial conditions the positive exponential term eventually dominates and x_1 becomes unbounded. This type of equilibrium point is called a **saddle point**.

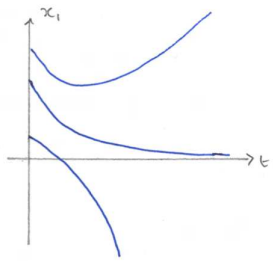


Figure 4.10: Examples of the solutions $x_1(t)$ for different initial conditions for the same case as in Figure 4.9.

From our previous examples, Example 4.11 is a saddle point.

4.5.3 Real and equal eigenvalues

Now we consider the case in which $r_1 = r_2 = r < 0$ (if $r > 0$ the trajectories in phase space are similar to what we describe below except the direction of motion is reversed). There are two subcases to consider depending on whether the repeated eigenvalue has two independent eigenvectors or just one.

1. If there are two independent eigenvectors then the solution has the form

$$\begin{aligned}\boldsymbol{x} = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} &= c_1\boldsymbol{\xi}^{(1)}e^{r_1 t} + c_2\boldsymbol{\xi}^{(2)}e^{r_2 t} \\ &= e^{rt}(c_1\boldsymbol{\xi}^{(1)} + c_2\boldsymbol{\xi}^{(2)})\,.\end{aligned}$$

Note that the ratio of x_1/x_2 is independent of t and so each trajectory is a straight line through the origin. Example trajectories are shown in Figure 4.11. This type of equilibrium point is called a **proper node** or **star point**. It is asymptotically stable (see the later definition 4.15) if $r < 0$ and unstable if $r > 0$.

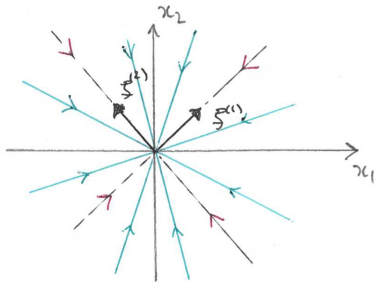


Figure 4.11: The phase portrait of an asymptotically stable proper node or star point with $r_1 = r_2 = r < 0$ and two independent eigenvectors.

2. When there is only one independent eigenvector the general solution takes the form (see §4.4.3)

$$\begin{aligned} \boldsymbol{x} &= c_1\boldsymbol{\xi}e^{rt} + c_2(\boldsymbol{\xi}te^{rt} + \boldsymbol{\eta}e^{rt})\,. \\ &= [(c_1\boldsymbol{\xi} + c_2\boldsymbol{\eta}) + c_2\boldsymbol{\xi}t]e^{rt}\,, \\ &= \boldsymbol{y}e^{rt}, \end{aligned} \tag{4.30}$$

say. As $t \to \infty$ the dominant term is $c_2\boldsymbol{\xi}te^{rt}$ and so at large times every trajectory approaches the origin tangent to $\boldsymbol{\xi}$. (Note that this is true even if $c_2 = 0$ because then the solution is $c_1\boldsymbol{\xi}e^{rt}$.) Similarly, if we also consider $t \to -\infty$ then again the term $c_2\boldsymbol{\xi}te^{rt}$ dominates. Hence the trajectory is parallel to the vector $\boldsymbol{\xi}$ far into the future and far into the past. An example of the phase portrait is shown in figure 4.12.

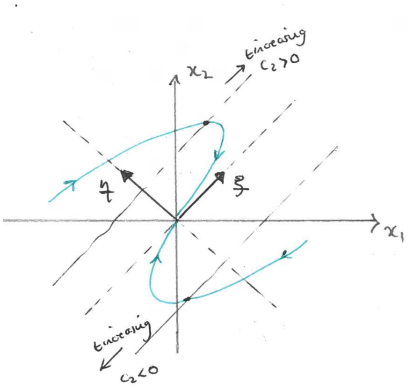


Figure 4.12: The phase portrait of an asymptotically stable improper node or degenerate point with $r_1 = r_2 = r < 0$ and only one independent eigenvector.

This type of equilibrium point is called a **improper node** or **degenerate node**. As for the proper node, it is asymptotically stable if $r < 0$ and unstable if $r > 0$.

From our previous examples, Example 4.14 is an unstable improper node.

4.5.4 Complex conjugate eigenvalues

First we consider the case in which the eigenvalues are $\lambda \pm i\mu$ (where λ and μ are real) with $\lambda \neq 0$ and $\mu > 0$. These systems typically take the form

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} \lambda & \mu \\ -\mu & \lambda \end{pmatrix} \boldsymbol{x} \tag{4.31}$$

which is equivalent to

$$x_1' = \lambda x_1 + \mu x_2, \quad x_2' = -\mu x_1 + \lambda x_2\,. \tag{4.32}$$

We can simplify this system by changing to polar coordinates r and θ given by

$$r^2 = x_1^2 + x_2^2, \quad \tan\theta = \frac{x_2}{x_1}\,.$$

Differentiating these expressions with respect to t gives

$$rr' = x_1x_1' + x_2x_2', \quad \theta' \sec^2\theta = \frac{x_1x_2' - x_2x_1'}{x_1^2}$$

and substituting for x_1' and x_2' from (4.32) gives

$$r' = \lambda r, \quad \theta' = -\mu\,,$$

since $\sec^2\theta = 1 + \tan^2\theta$. Integrating gives

$$r = ce^{\lambda t}, \quad \theta = -\mu t + \theta_0\,,$$

where c is a constant and $\theta_0 = \theta(t = 0)$. Thus we have parametric equations of the trajectories in polar coordinates. Since $\mu > 0$, θ decreases as t increases and the motion is clockwise in the $r\theta$-plane.

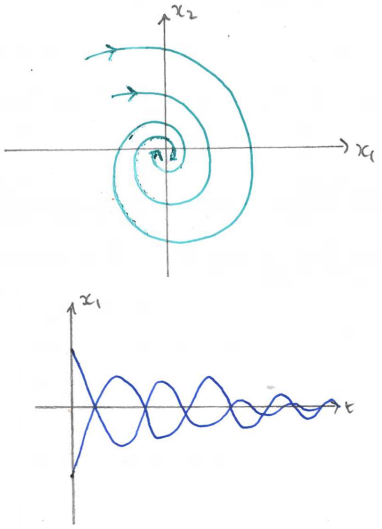


Figure 4.13: The phase portrait of an asymptotically stable spiral sink (top) and examples of the solutions $x_1(t)$ for different initial conditions (bottom).

As $t \to \infty$, $r \to 0$ if $\lambda < 0$ and $r \to \infty$ if $\lambda > 0$. Hence the trajectories are spirals which approach ($\lambda < 0$) or move away from ($\lambda > 0$) the origin. This type of equilibrium point is called an asymptotically stable **spiral sink** when $\lambda < 0$ (aka stable focus) or an unstable **spiral source** when $\lambda > 0$ (aka unstable focus). Examples of phase portraits and the solutions $x_1(t)$ are shown in figures 4.13 and figures 4.14.

From our previous examples, Example 4.13 is a spiral sink.

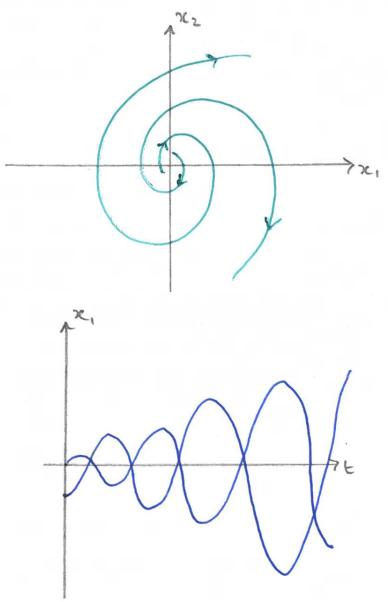


Figure 4.14: The phase portrait of an unstable stable spiral source (top) and examples of the solutions $x_1(t)$ for different initial conditions (bottom).

Pure imaginary eigenvalues

Note that when we have pure imaginary eigenvalues (i.e. $\lambda = 0$) the same analysis as above leads to

$$r = c, \quad \theta = -\mu t + \theta_0 \,,$$

where c and θ_0 are constants. Hence the trajectories are circles, with centre the origin. They are traversed clockwise if $\mu > 0$ and anti-clockwise if $\mu < 0$. The solutions are periodic with period $2\pi/\mu$. The equilibrium point is referred to as a **centre** and it is stable. An example phase portrait and some solutions $x_1(t)$ are shown in figure 4.15.

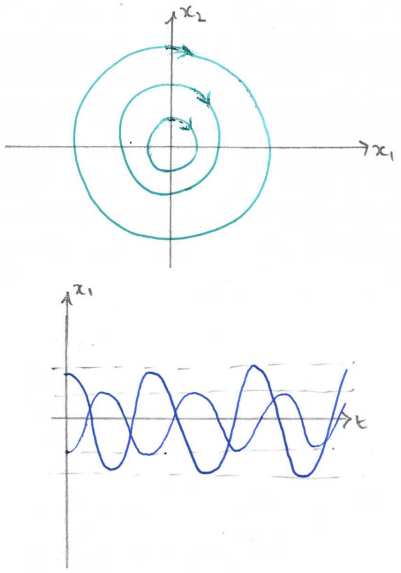


Figure 4.15: The phase portrait of a stable centre (top) and examples of the solutions $x_1(t)$ for different initial conditions (bottom).

In the above discussion, we have referred to stable and unstable equilibrium points based on the geometrical structure of their phase portrait. To be more precise, we have the following definition:

Definition 4.15 For the autonomous system

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{A}\boldsymbol{x}\,, \tag{4.33}$$

where $\boldsymbol{A}$ is a constant $n \times n$ matrix, any points where $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ are **equilibrium points**.

An equilibrium point $\boldsymbol{x}^*$ is said to be **stable** if, given any $\epsilon > 0$, there exists a $\delta > 0$ such that every solution $\boldsymbol{x}(t)$ of system (4.33) that at $t = 0$ satisfies

$$|\boldsymbol{x}(0) - \boldsymbol{x}^*| < \delta,$$

exists for all $t > 0$ and satisfies

$$|\boldsymbol{x}(t) - \boldsymbol{x}^*| < \epsilon,$$

for all $t \geq 0$. An equilibrium point that is not stable is said to be **unstable**.

An equilibrium point is said to be **asymptotically stable** if it is stable and if there exists a δ_0, with $0 < \delta_0 < \delta$, such that if a solution $\boldsymbol{x}(t)$ satisfies

$$|\boldsymbol{x}(0) - \boldsymbol{x}^*| < \delta_0,$$

then

$$\lim_{t\to\infty} \boldsymbol{x}(t) = \boldsymbol{x}^*\,. \tag{4.34}$$

Informally, stable solutions are those that start sufficiently close to $\boldsymbol{x}^*$ (within δ) and stay close (within ϵ). For asymptotic stability, solutions must not only stay close, but must eventually approach $\boldsymbol{x}^*$ as $t \to \infty$.

The above ideas of stability will be useful in modules such as MTH3011 (Nonlinear Systems and Control) and MTH3039 (Computational Nonlinear Dynamics) where one encounters **nonlinear systems**

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{f}(\boldsymbol{x})$$

which can be approximated by an **almost linear system**

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{g}(\boldsymbol{x})$$

in the neighbourhood of the equilibrium point. One is then interested in whether the trajectories of the linear system are good approximations to the behaviour of the nonlinear system. We will not pursue this further in this module.

4.6 Fundamental matrix

Definition 4.16 A set of n linearly independent solutions $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}, \ldots, \boldsymbol{x}^{(n)}$ of the system of n first order linear equations

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} \tag{4.35}$$

on the interval $I : \alpha < t < \beta$ is said to be a **fundamental set of solutions** on that interval.

Definition 4.17 A square matrix function $\boldsymbol{\Psi}(t)$

$$\boldsymbol{\Psi}(t) = \begin{pmatrix} x_1^{(1)}(t) & \ldots & x_1^{(n)}(t) \\ \vdots & & \vdots \\ x_n^{(1)}(t) & \ldots & x_n^{(n)}(t) \end{pmatrix} \tag{4.36}$$

whose columns are a fundamental set of solutions of system (4.35) is called a **fundamental matrix** of that system. Note that a fundamental matrix is nonsingular since its columns are linearly independent vectors.

Example 4.18 Find the fundamental matrix for the system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \boldsymbol{x}$$

4.6.1 General solution in terms of the fundamental matrix

When written in terms of the fundamental matrix $\mathbf{\Psi}(t)$, the general solution

$$\boldsymbol{x} = c_1\boldsymbol{x}^{(1)}(t) + \cdots + c_n\boldsymbol{x}^{(n)}(t)$$

of (4.35) becomes

$$\boldsymbol{x} = \mathbf{\Psi}(t)\boldsymbol{c}\,. \tag{4.37}$$

Here $\boldsymbol{c} = (c_1, \ldots, c_n)^T$ is the vector of arbitrary constants. For the initial value problem comprising the differential equation (4.35) and the initial conditions

$$\boldsymbol{x}(t_0) = \boldsymbol{k}\,, \tag{4.38}$$

where t_0 is a given point in the interval $I : \alpha < t < \beta$ and $\boldsymbol{k}$ is a given initial vector, we must choose $\boldsymbol{c}$ so that

$$\boldsymbol{x}(t_0) = \mathbf{\Psi}(t_0)\boldsymbol{c}\,.$$

Since $\mathbf{\Psi}(t_0)$ is nonsingular we have

$$\boldsymbol{c} = \mathbf{\Psi}(t_0)^{-1}\boldsymbol{x}(t_0)$$

and the solution of the initial value problem is then, from (4.37),

$$\boldsymbol{x} = \mathbf{\Psi}(t)\mathbf{\Psi}(t_0)^{-1}\boldsymbol{x}(t_0)\,. \tag{4.39}$$

Since each column of the fundamental matrix $\mathbf{\Psi}$ is a solution of (4.35), it follows that $\mathbf{\Psi}$ satisfies the matrix differential equation

$$\frac{d\mathbf{\Psi}}{dt} = \boldsymbol{P}(t)\mathbf{\Psi}\,. \tag{4.40}$$

Activity 4.19 Confirm (4.40) by comparing the jth columns of each side.

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4.6.2 Special form of the fundamental matrix

Theorem 4.20 Let

$$\boldsymbol{e}^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \boldsymbol{e}^{(2)} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \ldots \quad \boldsymbol{e}^{(n)} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}, \tag{4.41}$$

and let $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}$ be the solutions of system (4.35) that satisfy the initial conditions

$$\boldsymbol{x}^{(1)}(t_0) = \boldsymbol{e}^{(1)}, \quad \ldots, \quad \boldsymbol{x}^{(n)}(t_0) = \boldsymbol{e}^{(n)}, \tag{4.42}$$

where t_0 is any point in the interval $I : \alpha < t < \beta$. Then $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}$ form a fundamental set of solutions of system (4.35).

Sometimes it is useful to use the **special fundamental matrix**, that we will denote by $\boldsymbol{\Phi}(t)$, whose columns are the vectors $\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(n)}$ defined in Theorem 4.20. Then, at $t = t_0$ we have

$$\boldsymbol{\Phi}(t_0) = \begin{pmatrix} 1 & 0 & \ldots & 0 \\ 0 & 1 & \ldots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \ldots & 1 \end{pmatrix} = \boldsymbol{I}. \tag{4.43}$$

Now note that $\boldsymbol{\Phi}^{-1}(t_0) = \boldsymbol{I}$ and so from (4.39) we have

$$\boldsymbol{x} = \boldsymbol{\Phi}(t)\boldsymbol{x}(t_0). \tag{4.44}$$

Expression (4.44) represents a transformation of the initial condition $\boldsymbol{x}(t_0)$ into the solution $\boldsymbol{x}(t)$ at arbitrary time t. The advantage of (4.44) is that once the special fundamental matrix $\boldsymbol{\Phi}$ has been found, we can find the solution of a matrix system with many different initial conditions simply through matrix multiplication. Comparing (4.39) and (4.44) we see that

$$\boldsymbol{\Phi}(t) = \boldsymbol{\Psi}(t)\boldsymbol{\Psi}^{-1}(t_0). \tag{4.45}$$

Example 4.21 Find the special fundamental matrix $\boldsymbol{\Phi}(t)$ with $\boldsymbol{\Phi}(0) = \boldsymbol{I}$ for the system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \boldsymbol{x}\,.$$

.

4.7 Nonhomogeneous Linear Systems

We now consider the nonhomogeneous system

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} + \boldsymbol{g}(t)\,, \tag{4.46}$$

where the $n \times n$ matrix $\boldsymbol{P}(t)$ and the column vector $\boldsymbol{g}(t)$ are continuous on the interval $I : \alpha < t < \beta$. The general solution of (4.46) can be written in the form

$$\boldsymbol{x} = c_1\boldsymbol{x}^{(1)}(t) + \cdots + c_n\boldsymbol{x}^{(n)}(t) + \boldsymbol{w}(t)\,, \tag{4.47}$$

where $\boldsymbol{x}^{(1)}(t), \ldots, \boldsymbol{x}^{(n)}(t)$ is a fundamental set of solutions of the corresponding homogeneous problem

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} \tag{4.48}$$

and $\boldsymbol{w}(t)$ is a particular solution of the nonhomogeneous system (4.46). We will consider several methods of finding the particular solution $\boldsymbol{w}$.

4.7.1 Method of Undetermined Coefficients

This method is useful if the coefficient matrix $\boldsymbol{P}$ is a constant matrix ($= \boldsymbol{A}$, say) and if the components of $\boldsymbol{g}$ are polynomials, exponentials, sinusoidal functions or sums or products of these. The method follows that for second order ODEs, where we search for a particular integral of the same form as $\boldsymbol{g}$. One main difference is when $\boldsymbol{g}$ has the form $\boldsymbol{u}e^{\lambda t}$ and λ is a simple root of the characteristic equation. In this case, we might expect that we should seek solutions of the form $\boldsymbol{a}te^{\lambda t}$. However, it is necessary to use $\boldsymbol{a}te^{\lambda t} + \boldsymbol{b}e^{\lambda t}$. Then $\boldsymbol{a}$ and $\boldsymbol{b}$ are determined by substitution.

Example 4.22 Use the method of undetermined coefficients to find the general solution of

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}\boldsymbol{x} + \begin{pmatrix} 2e^{-t} \\ 3t \end{pmatrix} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{g}(t)$$

.

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4.7.2 Method of Diagonalisation

Consider the nonhomogeneous system

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{g}(t)\,, \tag{4.49}$$

where the $n \times n$ diagonalisable matrix $\boldsymbol{A}$ is a constant matrix. The method proceeds by defining a new variable $\boldsymbol{y}$ by

$$\boldsymbol{x} = \boldsymbol{T}\boldsymbol{y}\,,$$

where $\boldsymbol{T}$ is a matrix whose columns are the (constant) eigenvectors of $\boldsymbol{A}$. Then by substituting into (4.49) we obtain

$$\boldsymbol{T}\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{T}\boldsymbol{y} + \boldsymbol{g}(t)\,.$$

Now we multiply by $\boldsymbol{T}^{-1}$ and find

$$\boldsymbol{y}' = (\boldsymbol{T}^{-1}\boldsymbol{A}\boldsymbol{T})\boldsymbol{y} + \boldsymbol{T}^{-1}\boldsymbol{g}(t) = \boldsymbol{D}\boldsymbol{y} + \boldsymbol{h}(t)\,, \tag{4.50}$$

where $\boldsymbol{D}$ is the diagonal matrix whose diagonal entries are the eigenvalues $r_1, \ldots, r_n$ arranged in the same order as the corresponding eigenvectors $\boldsymbol{\xi}^{(1)}, \ldots, \boldsymbol{\xi}^{(n)}$ that form the columns of $\boldsymbol{T}$. System (4.50) is then a system of uncoupled linear ODEs for $y_1(t), \ldots, y_n(t)$, which can be solved by the methods of § 2.

Example 4.23 Find the general solution of the system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}\boldsymbol{x} + \begin{pmatrix} 2e^{-t} \\ 3t \end{pmatrix} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{g}(t)$$

using the method of diagonalisation.

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4.7.3 Variation of Parameters

Now we consider the more general nonhomogeneous problem

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} + \boldsymbol{g}(t), \tag{4.51}$$

with $\boldsymbol{P}(t)$ and $\boldsymbol{g}(t)$ continuous on the interval $I : \alpha < t < \beta$, in which the matrix $\boldsymbol{P}$ is not constant or not diagonalisable. We assume that the fundamental matrix for the corresponding homogeneous system

$$\frac{d\boldsymbol{x}}{dt} = \boldsymbol{P}(t)\boldsymbol{x} \tag{4.52}$$

has been found and we use the method of variation of parameters to find a particular solution (and hence the general solution) of (4.51).

We know that the general solution of the homogeneous problem (4.52) is $\boldsymbol{x} = \boldsymbol{\Psi}(t)\boldsymbol{c}$ (see (4.37)) and so, as we did for second order ODEs in §3.6, we seek a solution of the nonhomogeneous problem by replacing $\boldsymbol{c}$ by an unknown $\boldsymbol{u}(t)$, i.e. we set

$$\boldsymbol{x} = \boldsymbol{\Psi}(t)\boldsymbol{u}(t)\,,$$

where $\boldsymbol{u}(t)$ is to be found. Differentiating the product and substituting into (4.51) gives

$$\begin{aligned} \boldsymbol{\Psi}'(t)\boldsymbol{u}(t) + \boldsymbol{\Psi}(t)\boldsymbol{u}'(t) &= \boldsymbol{P}(t)\boldsymbol{\Psi}(t)\boldsymbol{u}(t) + \boldsymbol{g}(t)\,, \\ \Rightarrow \boldsymbol{\Psi}(t)\boldsymbol{u}'(t) &= \boldsymbol{g}(t)\,, \end{aligned}$$

because the fundamental matrix satisfies $\boldsymbol{\Psi}'(t) = \boldsymbol{P}(t)\boldsymbol{\Psi}(t)$ (see (4.40)). Now, $\boldsymbol{\Psi}(t)$ is nonsingular on any interval where $\boldsymbol{P}$ is continuous, so $\boldsymbol{\Psi}^{-1}(t)$ exists and we have

$$\begin{aligned} \boldsymbol{u}'(t) &= \boldsymbol{\Psi}^{-1}(t)\boldsymbol{g}(t) \\ \Rightarrow \boldsymbol{u}(t) &= \int \boldsymbol{\Psi}^{-1}(t)\boldsymbol{g}(t)dt + \boldsymbol{c}\,, \end{aligned} \tag{4.53}$$

where the constant vector $\boldsymbol{c}$ is arbitrary. Hence we find

$$\boldsymbol{x} = \boldsymbol{\Psi}(t)\boldsymbol{c} + \boldsymbol{\Psi}(t)\int \boldsymbol{\Psi}^{-1}(t)\boldsymbol{g}(t)dt\,. \tag{4.54}$$

Note that any initial condition can be satisfied by an appropriate choice for the arbitrary constant $\boldsymbol{c}$ and so (4.54) is the general solution. The first term on the rhs is the general solution of the homogeneous system (4.52) and the second term is a particular integral of (4.51).

For the initial conditions

$$\boldsymbol{x}(t_0) = \boldsymbol{k}$$

we can choose the particular solution in (4.54) as the one that is zero when $t = t_0$. This can be achieved by writing the general solution (4.54) in the form

$$\boldsymbol{x} = \boldsymbol{\Psi}(t)\boldsymbol{c} + \boldsymbol{\Psi}(t)\int_{t_0}^{t} \boldsymbol{\Psi}^{-1}(s)\boldsymbol{g}(s)ds\,. \tag{4.55}$$

and taking

$$\boldsymbol{c} = \boldsymbol{\Psi}^{-1}(t_0)\boldsymbol{x}(t_0)$$

so that the solution of the initial value problem is

$$\boldsymbol{x} = \boldsymbol{\Psi}(t)\boldsymbol{\Psi}^{-1}(t_0)\boldsymbol{x}(t_0) + \boldsymbol{\Psi}(t)\int_{t_0}^{t} \boldsymbol{\Psi}^{-1}(s)\boldsymbol{g}(s)ds\,. \tag{4.56}$$

Finally, note that if we use the special fundamental matrix $\boldsymbol{\Phi}(t)$ with $\boldsymbol{\Phi}(t_0) = \boldsymbol{I}$ then we have the simpler form

$$\boldsymbol{x} = \boldsymbol{\Phi}(t)\boldsymbol{x}(t_0) + \boldsymbol{\Phi}(t)\int_{t_0}^{t} \boldsymbol{\Phi}^{-1}(s)\boldsymbol{g}(s)ds\,. \tag{4.57}$$

Example 4.24 Find the general solution of the system

$$\frac{d\boldsymbol{x}}{dt} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}\boldsymbol{x} + \begin{pmatrix} 2e^{-t} \\ 3t \end{pmatrix} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{g}(t)$$

using the method of variation of parameters.

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